

Sample Solutions

While Loops Function Stack Practice

(This code, and sample solutions, are also posted to the class webpage.)

1.

```
def f(n: int):
    i = 0
    while i < n:
        print(i)
        i += 1

def main1():
    f(2)
    f(4)

main1()
```

main1

f

n = 2

i = 0, 1, 2

f

n = 4

i = 0, 1, 2, 3, 4

Printed output

0

1

0

1

2

3

2.

```
def g1(n: int) -> int:
    if n > 3:
        return n + 1
    else:
        return n - 3
```

```
def g2(n: int):
    i = 2
    while i < n:
        print(g1(i))
        i += 1
```

```
def main2():
    g2(1)
    g2(4)
    g2(5)
```

```
main2()
```

main2

g2

n = 1

i = 2

(since i < n,
nothing happens.)

g2

n = 4

i = 2, 3, 4

g1

n = 2

F-1

g1

n = 3

F0

g2

n = 5

i = 2, 3, 4, 5

g1

n = 2

F-1

g1

n = 3

F0

g1

n = 4

F5

Printed output

-1

0

-1

0

5

3.

```
def h(a: int, b: int):  
    count = 0  
    while a < b:  
        i = 0  
        while i < a:  
            count += i  
            i += 1  
  
        a *= 2  
  
    return count  
  
def main3():  
    x = h(2,7)  
    print(x)  
  
main3()
```

main 3
x = 7

h
a = 2 4 8

b = 7
count = 0 0 1 1 2 4 7

i = 0 1 2 3 4

7

printed output
7